

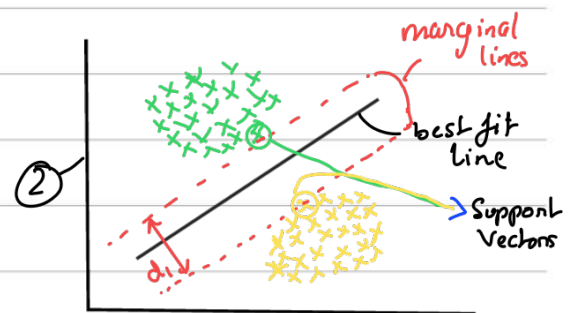
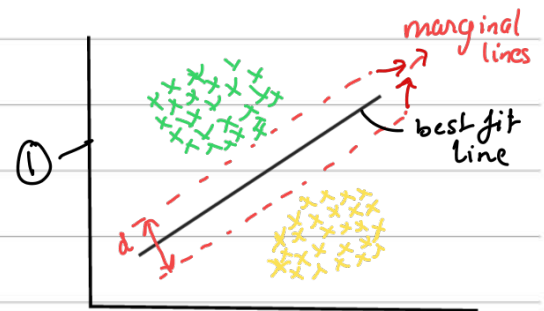
Support Vector Machine ML Algorithm:

- 1- SVC (Support Vector Classifier): $\rightarrow$ classification
- 2- SVR (Support Vector Regressor): $\rightarrow$ Regression

I)- Support Vector classifier: (SVC)

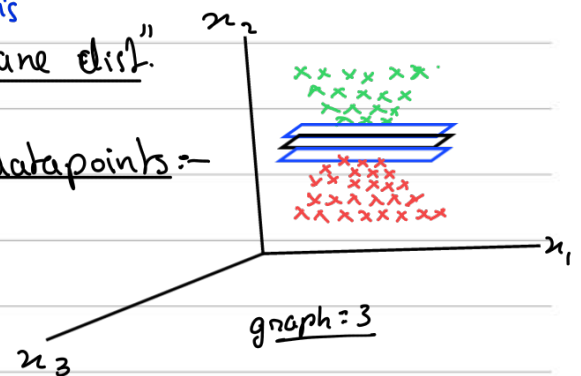
SVM algorithm also creates 2 parallel marginal lines along with the best fit line (also ca "Marginal planes" if dimension is more than 2.)

also the distance b/w the marginal lines must be maximum such that they should pass through the nearest points to the best fit line on both sides



★ Distance d_1 should be maximum
i.e. $d_1 > d$, where the distance is
maximum is called "Marginal Plane dist."

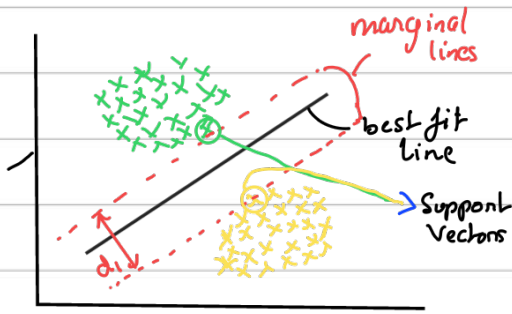
Visualizing marginal plane for 3D datapoints:-



But,

in most of the dataset such clean separation b/w datapoints
as in graph ① & ② are not available

1.1)- Soft Margin and Hard Margin in SVC:

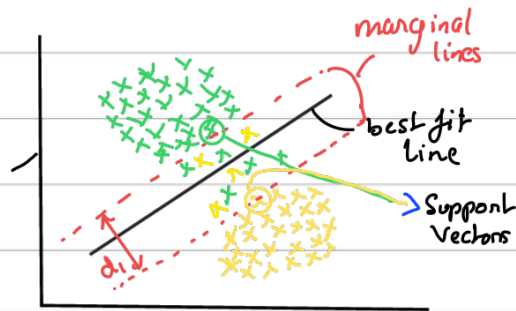


In this scenario we can easily segregate the datapoints very clearly and $\text{Error} = 0$

∴ ① **Hard Margin** = Non of the datapoints are misclassified
(impossible scenario)

But, getting Hard margin in a real life dataset is almost impossible to achieve.

So, in such cases the Best fit line and marginal lines are created with some errors. ex:-



No. of errors are less compared to completely classified line.

∴ ② **Soft Margin**:- Some of the data is misclassified, But majority is separated.

1.2)- How to create Best-fit and Marginal Line/Plane: (24:35) cont

eqⁿ for best fit line:

$$y = mx + c \quad \text{or} \quad ax + by + c = 0 \quad \text{or} \quad h\theta(x) = \theta_0 + \theta_1 x_1$$
$$\Rightarrow y = \underbrace{\left(-\frac{a}{b}\right)}_m x - \underbrace{\left(\frac{c}{b}\right)}_c$$

also for multi-feature linear regression we know,

$$y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 \text{ (for 3 i/p's)}$$

can also be written as,

$$y = b + [\omega_1 x_1 + \omega_2 x_2 + \omega_3 x_3] \quad , \quad b \rightarrow \text{slope} \quad \& \quad \omega_i \rightarrow \text{coeff}$$

representing using matrix,

$$\omega = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} \quad \& \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

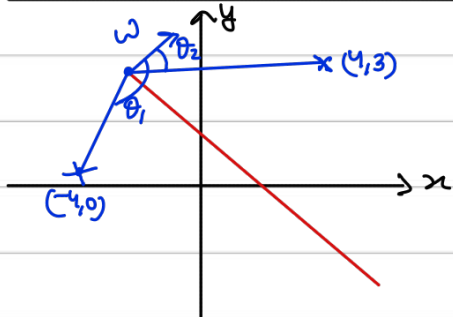
$$\omega \cdot x = \omega^T x = \begin{bmatrix} \omega_1 & \omega_2 & \omega_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\Rightarrow \boxed{y = \omega^T x + b} \rightarrow \text{similar to } y = mx + c \text{ or } ax + by + c = 0$$

Since $y = \omega^T x + b$ is similar to $ax + by + c = 0 \therefore$,
we get,

$$\star \boxed{\omega^T x + b = 0} \text{ --- ① } \underline{\text{Straight Line eqn}}$$

+ve and -ve distance:-



Since $\angle \theta_1 > 90 \Rightarrow$ -ve distance and
 $\angle \theta_2 < 90 \Rightarrow$ +ve distance above the
normal line.

Marginal Plane in SVM:-



$$\text{let } P_1 \in x_1 \Rightarrow (x_1, x_2)$$

$$P_2 \in x_2 \Rightarrow (x_1, x_2)$$

$$\left. \begin{array}{l} \omega^T x + b = +1 \text{ --- ①} \\ \omega^T x + b = 0 \text{ --- ②} \\ \omega^T x + b = -1 \text{ --- ③} \end{array} \right\} \begin{array}{l} \text{subtracting eq ① \& ③} \\ \omega^T x + b = +1 \\ \omega^T x + b = -1 \\ \hline \omega^T (x_1 - x_2) = 2 \text{ --- ④} \end{array}$$

on dividing eq ③ with magnitude of w i.e. $\|w\|$

$$\Rightarrow \frac{\omega^T (x_1 - x_2)}{\|\omega\|} = \frac{2}{\|\omega\|} \text{ --- distance b/w marginal planes.}$$

Cost Function:- maximize $\frac{2}{\|w\|}$ i.e. the distance b/w Marginal plane.

Constraints:-
$$y_i = \begin{cases} +1 & \text{if } w^T x + b \geq 1 \\ -1 & \text{if } w^T x + b \leq -1 \end{cases}$$

$\therefore$

For all correctly classified datapoints the ϕ will always be $\boxed{y_i * w^T x + b \geq 1}$

Modified Cost Funⁿ of SVC:-

We know, maximize (w, b) :- $2/\|w\|$

$\Rightarrow$ minimize (w, b) :- $\frac{\|w\|}{2}$

and constraints will be:-

$$y_i = \begin{cases} +1 & \text{if } w^T x + b \geq 1 \\ -1 & \text{if } w^T x + b \leq -1 \end{cases}$$

Cost function of Soft Margin

Now to handle the cost funⁿ when there are overlapped data points. $\therefore$ we

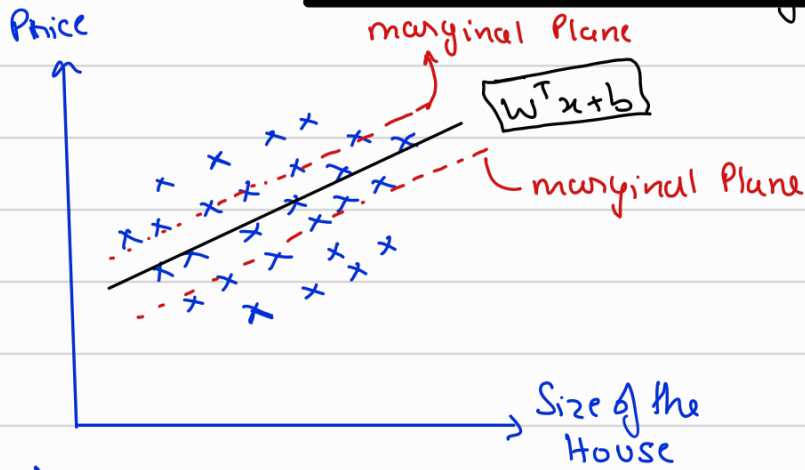
$$\text{Cost Fun}^n = \min_{(w, b)} \frac{\|w\|}{2} + C_i \sum_{i=1}^n \xi_i$$
 "Hinge Loss" (Hyperparameter)



$C_i \rightarrow$ How many points we can consider for misclassification.

$\sum_{i=1}^n \xi_i \rightarrow$ Summation of distance of incorrect data-points to the marginal planes of opposite side

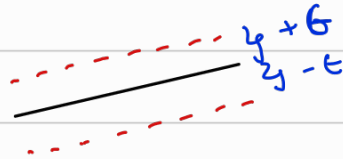
Support Vector Regression (SVR)



Now,

Suppose if the prediction is done w.r.t. ^{above} marginal plane, on best fit line then there will be some difference in the prediction thus, marginal errors (ϵ) is introduced

Marginal Error (ϵ):- dist b/w the above plane to the best fit line



Cost Funⁿ:-

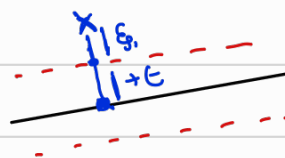
$$\min_{w, b} \frac{\|w\|}{2} + \boxed{C \sum_{i=1}^n \xi_i} \rightarrow \text{Hinge loss}$$

Constraint:-

$$\underbrace{|y_i - w^T x_i|}_{\text{the distance b/w best fit line and actual data point}} \leq \epsilon + \underbrace{\xi_i}_{\text{error above the margin such that}}$$

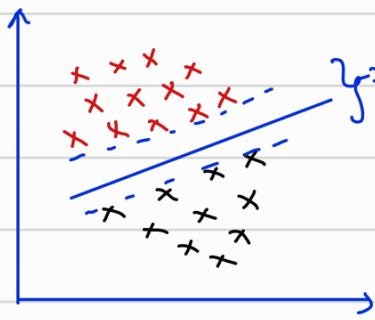
the distance b/w best fit line and actual data point $\leq \epsilon$

Since all data points can't fall b/w the marginal planes and best fit line thus we have $\epsilon + \xi_i$ as $\rightarrow$ Error above the margin such that



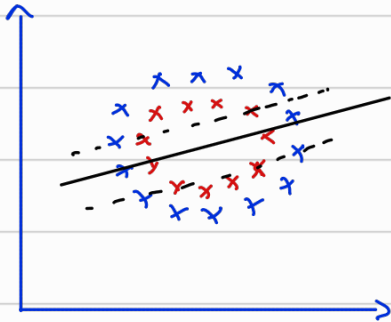
$\xi_i + \epsilon$ is the max dist b/w best fit line and actual data point.

SVM Kernel



→ 2 marginal lines and 1 best fit line

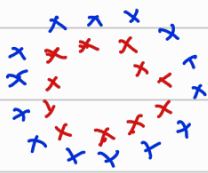
we know whenever we use straight best fit and marginal lines we call it "Linear SVC"



→ accuracy = 50%

cuz some of data points are outside the the marginal plane

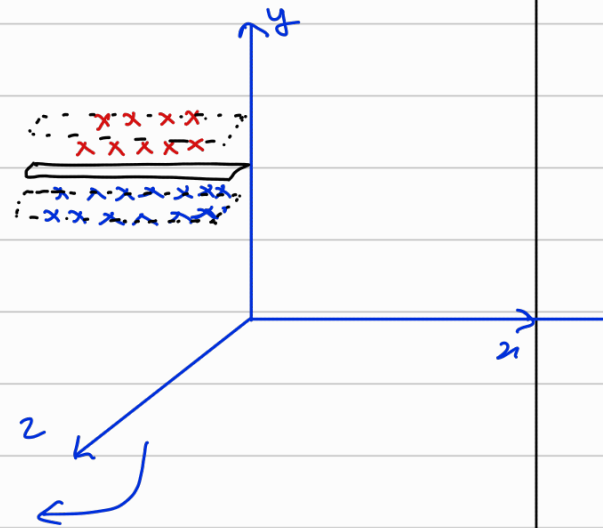
Can't use Linear SVC in this case,
we use "SVM kernels".



⇒ Transformation ⇒

using some
mathematical
formulae

2D → 3D



Now after conversion 2D is 3D using
some mathematical formulae we can
apply "Linear SVC" on 3D plane

ex Dataset

x	y
2	Yes
3	No
4	Yes
-	No
-	-

$\Rightarrow$

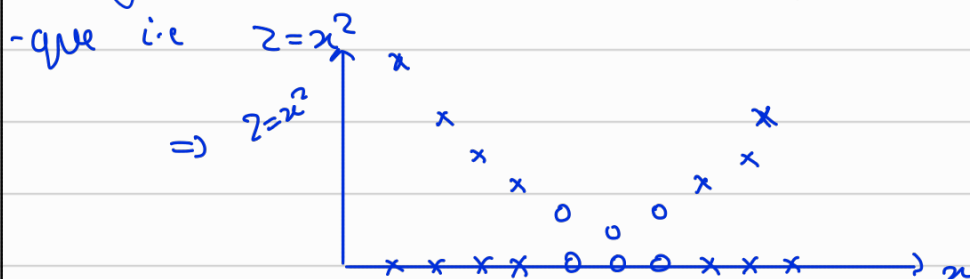
0 - No
x - Yes

$\rightarrow$ $x \times x \times x \times 0 \times 0 \times x \times x \times$

can't apply Linear SVC
 coz: if best fit line
 is created on one side
 rest will become error

now,

using SVM kernel we will apply data transformation techni-



Types of SVC Kernels:-

- 1) Polynomial Kernel
 - 2) RBF Kernel
 - 3) Sigmoid Kernel
- } Transformation

1) Polynomial Kernel:-

$$f(x, y) = (x^T y + c)^d, \quad c = 1$$

$d = \text{no. of features}$

$$= \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2 \times 1} \begin{bmatrix} x_1 & x_2 \end{bmatrix}_{1 \times 2}$$

$$x = [x_1, x_2] \text{ \& \; } y = [x_1, x_2]$$

$$= \begin{bmatrix} x_1^2 & x_1 x_2 \\ x_2 x_1 & x_2^2 \end{bmatrix}$$

previously we had only 2 features (x_1 and x_2) but now polynomial kernel has created 3 new features (x_1^2 , $x_1 x_2$ & x_2^2).



2y RBF kernel:-